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# SIH26151 â€” Final Build Guide
## Evidence-First Dark-Web Threat Actor Attribution Assistance Platform

**Document status:** Authoritative implementation guide
**Audience:** Human engineers, technical leads, coding agents and SIH team members
**Baseline repository:** `https://github.com/Sharan112212/darkweb`
**Build model:** Extend existing modules; add adapters and shared contracts; do not perform
a generic rewrite.
**Primary requirement:** Every displayed association must be evidence-backed, confidence-
scored, caveated and reviewable.

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# Part A â€” Product Definition

## 1. Mission

Build an analyst-operated platform that correlates authorized dark-web and related
technical observations into candidate actor associations. It may reveal technical
overlap between aliases, keys, wallets, services, infrastructure and writing patterns; it
must never claim to reveal a natural personâ€™s identity or to defeat Tor.

Mandatory product disclosure, visible in the dashboard and reports:

> This system provides confidence-scored technical associations for authorized analyst
review. It does not defeat Tor, establish a personâ€™s real-world identity, or replace
legal/forensic investigation. Confidence weights are configurable prototype methodology, not
a validated intelligence standard.

## 2. Problem outcome

An analyst must be able to:

1. Search a handle, PGP fingerprint, wallet, contact, onion hostname or domain.
2. Inspect all observations and their provenance.
3. See related entities in a time-bounded graph.
4. Inspect every edgeâ€™s evidence chain and caveats.
5. Review a candidate association and accept, reject or defer it.
6. Detect a possible persona migration/rebrand.
7. Export an auditable case report.
8. Run the full demo in offline synthetic-fixture mode.

## 3. Scope boundary

### Allowed

- Synthetic fixtures and explicitly authorized public sources.
- Passive collection of allowlisted content using configured Tor proxy isolation.
- Metadata, text and explicitly observable technical indicators.
- Analyst review and evidence-based correlation.

### Prohibited

- Exploiting sites, submitting forms, logging in, bypassing CAPTCHA/access controls,
purchasing goods, interacting with marketplaces or attempting to find a userâ€™s IP.
- Executing untrusted JavaScript, opening hostile files automatically or analyzing malware
in the collector runtime.
- Claiming real-world identity, criminal culpability or conclusive de-anonymization.

## 4. User roles

| Role | Can do | Cannot do |
|---|---|---|
| Viewer | Search redacted results, view accepted/reviewable links | View raw protected

```

evidence, export sensitive reports, decide links |
| Analyst | Search, open evidence, create cases, propose/accept/reject links with reason,
export allowed cases | Change source policy or system scoring configuration |
| Reviewer | All analyst actions; approve high-tier associations where configured | Modify
collection policy |
| Admin | Manage users, sources, limits, redaction rules, deployment configuration | Bypass
audit logging |

Part B “ Existing Repo and Open-Source Plan

5. Baseline rule

Before implementation, audit the baseline repository. The build assumes it already includes a Tor scraper/collector, `infra-matcher/match\_infra.py`, identity graph, `stylometry.py` using MiniLM, `fusion.py`, a dashboard and Docker deployment. Do not replace any of these until Branch 0 has verified their exact paths, inputs, outputs and dependencies.

6. Open-source decision register

Use directly, without modifying upstream source

Tool/component	Purpose	How it is used
Existing repo collector	Source collection	Retain and harden through configuration/wrappers
Existing infra matcher	Certificate/twin correlation	Keep native matching logic; emit evidence through adapter
Existing identity graph	Hard identifier linking	Keep current resolver; adapt output to canonical schema
Existing MiniLM module	Semantic similarity	Keep as a supporting text signal only
`nao1215/onionscan`	Onion OPSEC/misconfiguration observations	Pinned binary/container invoked as restricted subprocess; no fork
`pystylometry`	Classical writing-style features	Optional Python dependency invoked through wrapper
PostgreSQL	Transactional metadata, cases, audit, jobs	Docker service, unmodified
MinIO	Immutable raw artifact storage	Docker service, unmodified
Neo4j Community Edition	Stretch graph backend	Optional, via a repository adapter, unmodified

Adapt locally; do not change upstream project code

Local adapter	Input	Output	Why
`infra_evidence_adapter.py`	Existing infra matcher result	Validated `EvidenceUnit`	Preserve existing logic and create provenance-rich output
`identity_evidence_adapter.py`	Existing identity graph result	Validated `EvidenceUnit`	Keep custom resolution but standardize evidence
`minilm_evidence_adapter.py`	Existing `stylometry.py` result	Validated `EvidenceUnit`	Label semantic similarity accurately and cap its effect
`onionscan_adapter.py`	Saved/raw OnionScan result	Validated `EvidenceUnit`	Avoid forking scanner; retain tool output for audit
`classical_stylometry.py`	Eligible post corpora	Validated `EvidenceUnit`	Enforce corpus/language gates and calibration
`behavior_engine.py`	Alias post histories	Validated `EvidenceUnit`	Generate migration signals from timing/template/vocabulary

Do not integrate in MVP

Do not add	Reason
SpiderFoot, Maltego, Recon-ng	Generic OSINT breadth distracts from focused attribution; adds integration and licensing/maintenance risk
Full transaction-graph tracing (Chainalysis/GraphSense scope)	Requires independent data,

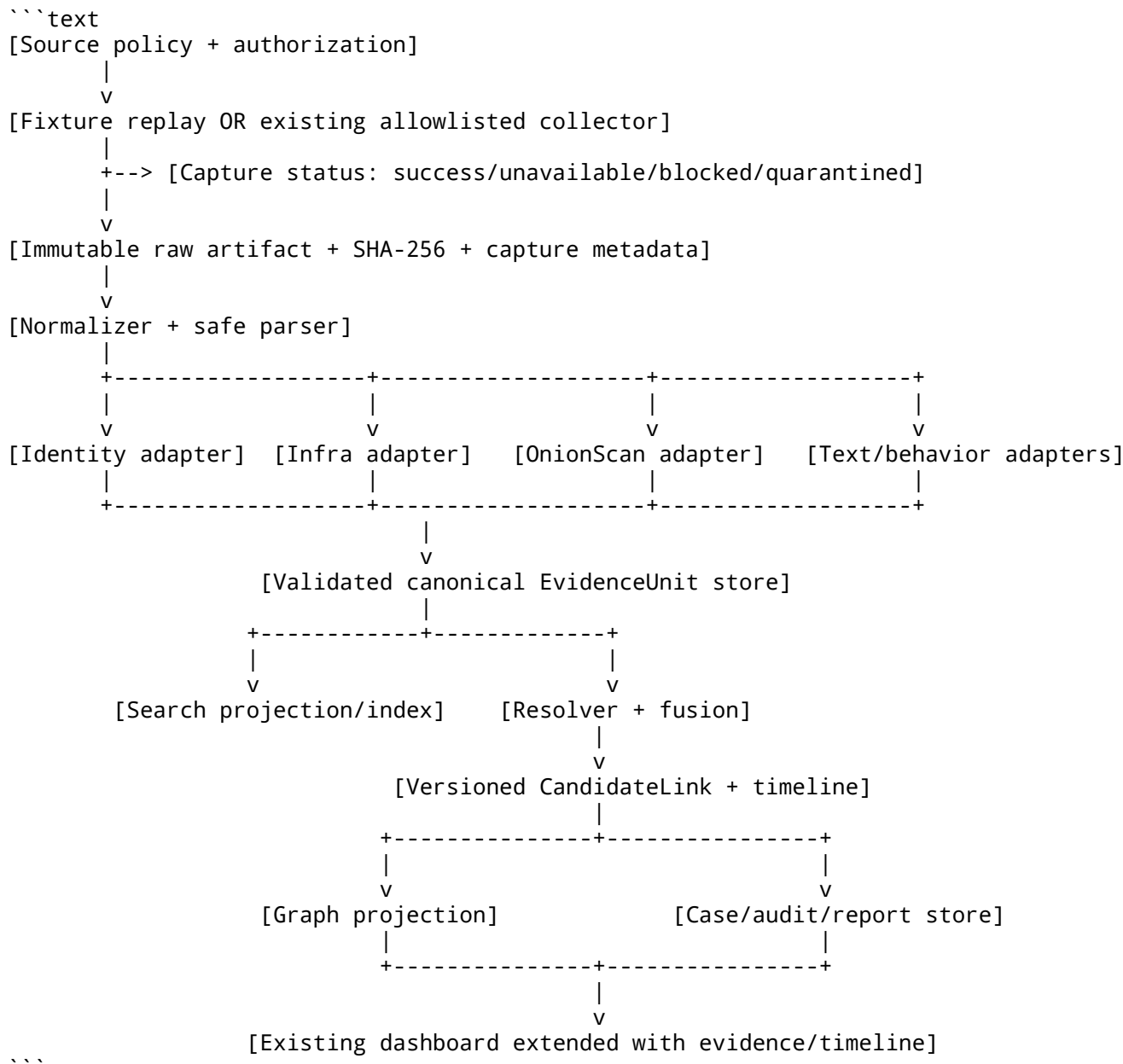
specialist heuristics and a separate product roadmap |
 | Kafka, Kubernetes, KubeFlow | Scale infrastructure is not required for a reliable single-node SIH demo |
 | Legacy original OnionScan codebase | Avoid old/unmaintained baseline; use current supported upstream package/tooling through wrapper |

7. Dependency governance

Create and maintain `docs/third-party-notices.md` with name, URL, exact version/commit, license, purpose, integration mode, whether upstream code was modified and update owner. Pin Python dependencies in a lock file, pin Docker images by digest where practical, create an SBOM, scan dependencies in CI and keep an offline artifact manifest for demo deployment.

Part C – Architecture and Data Contracts

8. Architecture



9. System-of-record rules

| Data | Canonical system | Rebuildable projection |

```

|---|---|---|
| Raw capture/scanner output | MinIO or immutable fixture archive | No |
| Capture metadata, source policy, evidence records, jobs, cases, decisions, audit |
PostgreSQL | No |
| Searchable redacted text/entities | PostgreSQL-backed data then OpenSearch if used | Yes |
| Graph nodes/edges | PostgreSQL/canonical link records first; Neo4j after migration | Yes |
| Reports | Immutable export artifact + export manifest in PostgreSQL | No |

```

Search and graph projections must never be treated as the only source of truth. They must be reconstructable from canonical evidence/link records.

10. Canonical `EvidenceUnit`

All modules must emit exactly this logical contract. No module-specific payload may enter fusion directly.

```

```python
class EvidenceUnit:
 evidence_id: str
 schema_version: str
 capture_id: str

 source: str
 source_version: str
 indicator_type: str
 indicator_value: str
 indicator_role: str | None
 linked_entities: list[str] # exactly two when used for a candidate link

 confidence_weight: float # intrinsic signal strength, 0..1
 source_reliability: float # 0..1
 extraction_confidence: float # 0..1

 source_claimed_time: str | None
 observation_date: str | None
 captured_at: str
 time_confidence: float # 0..1

 source_url: str
 raw_evidence_hash: str
 raw_evidence_reference: str
 independence_group_id: str
 collector_mode: str #
 fixture_replay/authorized_tor/authorized_cleynet
 processing_status: str #
 valid/quarantined/parse_failed/redacted/superseded

 explanation: str
 limitations: list[str]
 context_excerpt: str | None
 model_metadata: dict
...

```

### ### Required semantic rules

- `evidence\_id`, `capture\_id`, `raw\_evidence\_hash`, `source\_url` and `captured\_at` are mandatory for valid evidence.
- Preserve the original source time separately from system capture time.
- If date is absent or unreliable, `observation\_date=None` and `time\_confidence` must be low.
- `independence\_group\_id` groups duplicated/reposted observations so they cannot inflate confidence.
- `indicator\_role` distinguishes `key\_published`, `verified\_signature`, `wallet\_unknown`, `shared\_service\_wallet`, `mixer\_suspected`, and similar contexts.
- Evidence cannot be silently changed. Corrections create a new superseding

version/reference.

## 11. Canonical `CandidateLink`

```
```python
class CandidateLink:
    link_id: str
    link_version: int
    left_entity_id: str
    right_entity_id: str

    state: str #
    proposed/accepted/rejected/needs_review/superseded
    score: float
    tier: str #
    insufficient_evidence/unresolved/possible_association/likely_same_actor/observed_technical_i
    dentity
    score_status: str # observed/insufficient/conflicting/stale
    category_breakdown: dict # K/I/B/S score + state + evidence IDs

    evidence_ids: list[str]
    conflict_set_id: str | None
    competing_link_ids: list[str]
    explanation: str
    limitations: list[str]

    score_model_version: str
    calculation_input_hash: str
    created_at: str
    updated_at: str
...

```

No actor merge is destructive. Store association memberships and analyst decisions as versioned relationships, so a later decision can split a wrongly consolidated candidate actor.

12. Normalization rules

Data type	Required normalization	Caution
PGP fingerprint	Uppercase, remove spaces, validate hex	A published key is not proof of key control; verified signature is stronger
Wallet	Validate supported chain format; retain chain/network	Shared escrow/mixer/donation addresses must be labeled and downweighted
Alias	Preserve display form; case-fold/trim for search	Detect Unicode confusables; do not automatically merge spelling variants
Contacts	Normalize by platform type	Redact in UI/export unless role permits visibility
Onion hostname	Lowercase, strip protocol/path; validate expected form	Do not infer physical location/operator
Domain	Lowercase/punycode; store registrable domain separately	Shared hosts/CDNs can cause false infrastructure links
Date	UTC ISO-8601 plus original timestamp	Do not trust source time without confidence

Part D – Safety, Edge Cases and Scoring

13. Mandatory collection controls

1. Allowlist source before network request; blocklist always overrides allowlist.
2. Store authorization status, approver and expiry for every live source.
3. Provide a global kill switch that stops new live jobs.
4. Force collector traffic through configured `socks5h` Tor endpoint; block direct internet

egress from collector container.

5. Per-host concurrency: 1. Configure request delay, timeout, response-size maximum and retry count.

6. Never execute JavaScript, submit forms, authenticate, solve CAPTCHA, download/extract archives automatically or follow cross-policy redirects.

7. MIME-allowlist content before parsing; quarantine malformed/binary/oversized data.

8. Retain status events for offline/unavailable/moved/blocked sources; never delete prior evidence because a source disappears.

9. Run parsers with bounded CPU/time/memory and isolate untrusted content handling.

14. Mandatory attribution controls

1. PGP key reuse, wallet reuse, alias reuse, infrastructure reuse and text similarity create candidate evidence, not real-world identity claims.

2. Exact alias alone cannot exceed `Possible Association`.

3. Infrastructure-only evidence cannot exceed `Possible Association` unless corroborated by independent hard evidence.

4. MiniLM and classical stylometry combined cannot exceed `Possible Association` without a non-text contextual signal.

5. Classical stylometry max total contribution to final score: 0.20.

6. Treat shared/mixer/escrow wallets as low-confidence contextual evidence unless independently supported.

7. Support conflict sets: where evidence supports competing hypotheses, show all relevant candidates rather than silently selecting one.

8. Flag one-off/low-reliability indicators as possible planted/deceptive indicators.

9. Do not destructively merge candidate actors; decisions must be reversible.

15. Scoring model

Use configurable values in `config/scoring.yaml`:

```
```text
```

$C = 0.35 * K + 0.25 * I + 0.20 * B + 0.20 * S$

K = hard identifiers/cryptographic evidence

I = infrastructure evidence

B = behavioral evidence

S = semantic + classical stylometric evidence

```
```
```

Calculate each category using deduplicated noisy-OR aggregation:

```
```text
```

```
adjusted_evidence = base_strength
 * source_reliability
 * extraction_confidence
 * freshness_factor
 * independence_factor
```

```
category_score = 1 - product(1 - adjusted_evidence_i)
```

```
```
```

For every category, return both a numerical value and a state: `observed`, `not\_available`, `insufficient`, `conflicting`, or `stale`. Absence of evidence is not evidence of absence.

Tier mapping

| Final score | Tier | Default treatment |
|-------------|-----------------------------|---|
| 0.00â€“0.19 | Insufficient Evidence | Hidden from default graph; remains searchable |
| 0.20â€“0.44 | Unresolved | Neutral relationship; no merge suggestion |
| 0.45â€“0.64 | Possible Association | Dotted edge; analyst review required |
| 0.65â€“0.84 | Likely Same Actor | Amber edge; review strongly recommended |
| 0.85â€“1.00 | Observed Technical Identity | Green edge; technical overlap only, never person-level identity |

Score stability

- Round display scores to two decimals.
- Persist the model version, configuration hash, corpus hash and input-evidence hash.
- When score/tier changes, create a score-change timeline event explaining whether the cause was new evidence, evidence removal/redaction, new configuration or model change.
- Use hysteresis near tier boundaries: a link must cross a boundary by at least 0.03 before automatic tier change, unless an analyst explicitly rescored/reviewed it.

16. Stylometry controls

Run classical stylometry only if each alias corpus has at least 5 posts, 1,500 cleaned characters and adequate same-language confidence. Remove quoted text, PGP blocks, wallet strings, repeated marketplace boilerplate and known templates before calculating style features.

Use MiniLM as **semantic similarity**, not authorship proof. Use classical metrics—function-word frequencies, sentence-length distribution, punctuation habits and character “5 grams” as a separate supporting signal. If language is mixed, translation is suspected, corpus is short or template dominance is high, emit no style evidence rather than an unreliable score.

17. Edge-case behavior table

| Condition | Required behavior |
|-------------------------------------|---|
| Source unavailable/timeout/503 | Create a source-status event; retain old evidence; bounded retry/backoff |
| Same content via mirror/repost | Retain observation, assign same independence group, count once for score |
| Malformed/oversized/binary response | Quarantine; persist capture metadata; do not parse/display raw content |
| CAPTCHA/login/JS-only page | Record `not_collected_reason`; do not bypass/automate |
| OnionScan failure/schema change | Preserve raw output/error; mark adapter failure; use fixture fallback in demo |
| Shared certificate/header/CDN | Downweight; caveat; never strong alone |
| Shared/mixer wallet | Label role; cap contribution; require independent corroboration |
| Shared/revoked/fake PGP key | Distinguish published key from verified signature; caveat |
| Unicode aliases/recycled handles | Preserve original and normalized form; calculate collision/rarity; no automatic merge |
| Evidence supports competing actors | Create conflict set and show competing hypotheses |
| Later evidence disproves link | Add superseding/rejection decision; preserve prior version |
| Forged/missing source time | Use captured time; lower time confidence; do not infer ordering |
| Model/config change | Version and rescore; preserve earlier output and explain change |
| Concurrent duplicate job | Idempotency keys + unique constraints; one canonical evidence record |
| Export during change | Snapshot exact evidence/link versions/config before rendering |
| Retention/redaction request | Apply authorized tombstone/redaction; preserve audit notice and report impact |
| Search returns nothing | Explain whether data was not collected, source unavailable, no match or result restricted |

Part E “ App Flow and Data Flow

18. Application flow

```
```text
[Login]
 -> [Search]
 -> [Evidence results]
```

```

-> [Entity detail]
 -> [Candidate actor profile]
 -> [Graph explorer]
 -> [Edge evidence drawer]
 -> [Accept / reject / needs review]
 -> [Timeline]
 -> [Timeline event evidence]
 -> [Add to case]
 -> [Case notes / decision history]
 -> [Export JSON / CSV / PDF]
-> [Admin source policy]
-> [Audit viewer]
...

```

## ## 19. UI requirements

### ### Search

Input types: alias, PGP fingerprint, wallet, contact, onion hostname, domain. Filters: source, date range, entity type, candidate tier and rejected-link visibility. Results show redacted excerpt, source, observation/capture date and evidence availability state.

### ### Candidate actor profile

Show tier, rounded score, disclaimer, aliases, indicators, sources, known limitations, graph/timeline/case tabs. Do not show "identity found" language.

### ### Graph explorer

Default to two hops. Use progressive expansion and node/edge limits. Edge styles: dotted grey for possible, neutral for unresolved, amber for likely, green for observed technical identity, hidden by default for rejected. Clicking an edge must open evidence; edge color alone is never sufficient explanation.

### ### Evidence drawer

Display exact chain:

```

```text
module/source -> observed indicator -> linked entities -> dates -> reliability -> confidence
contribution -> caveat
```

```

Show evidence IDs, capture hash, source URL/reference, model/rule version, raw evidence availability status and analyst decisions. Raw captures are not shown without permission.

### ### Timeline

Event types: first seen, last seen, post observed, alias observed, PGP/wallet/contact seen, infrastructure observation, candidate link created, score change, analyst decision, export. Date filter must apply consistently to graph and timeline.

### ### Case and export

Cases contain selected actors, links, evidence references, analyst notes and decision history. Export must make a snapshot before rendering; include report ID, exact evidence/link versions, score config/model version, timestamp, generated-by user and limitations.

## ## 20. End-to-end data flow

```

```text
1. Source policy checks fixture or approved source.
2. Collector replays/fetches content and creates Capture record.
3. Raw artifact is hashed and stored; status is recorded even when collection fails.

```


4. Normalizer validates MIME/size and produces safe text/metadata, or quarantines capture.
5. Each existing/new adapter emits validated EvidenceUnit records.
6. Evidence records persist to canonical store and become search/graph projections.
7. Resolver groups evidence by entity pair and independence group.
8. Fusion calculates K/I/B/S, final score, tier, limitations and explanation.
9. CandidateLink version is written; conflict sets and timeline events are updated.
10. Dashboard requests redacted data through APIs.
11. Analyst decision writes a new immutable audit event and relationship version.
12. Export snapshots selected versions and creates report artifact/audit record.

21. API contract minimum

```text

```
GET /health
POST /v1/search
GET /v1/entities/{type}/{id}
GET /v1/evidence/{evidence_id}
GET /v1/actors/{actor_id}
GET /v1/actors/{actor_id}/graph?depth=2&from=...&to=...
GET /v1/actors/{actor_id}/timeline?from=...&to=...
GET /v1/links/{link_id}
GET /v1/links/{link_id}/explanation
POST /v1/links/{link_id}/decision
POST /v1/cases
GET /v1/cases/{case_id}
POST /v1/exports
GET /v1/audit
GET /v1/admin/sources
POST /v1/admin/sources
```

```

Part F â€” Branch-by-Branch Build Plan

22. Universal branch rules

Before merging any branch:

1. State scope and explicit non-scope.
2. List changed data contracts and migration/rollback steps.
3. Add/update fixtures.
4. Add unit and integration tests covering happy and negative case.
5. Review source access, secrets, raw-data exposure and logs.
6. Update documentation, dependency notices and demo script as applicable.
7. Run lint/tests from clean Docker startup.
8. Attach acceptance proof: test output, API payload or screenshot.

Branch 0 â€” Baseline audit

****Goal:**** Verify the repositoryâ€™s actual implementation before change.

****Steps:**** Inspect collector, infra matcher, identity graph, `stylometry.py`, `fusion.py`, dashboard and Docker. Record file paths, language, entry point, inputs, outputs, dependencies, current test data, security limitations and mapping to `EvidenceUnit`.

****Data flow:**** Existing module output -> audit table only. No functional changes.

****Deliverables:**** `docs/baseline-audit.md`, module inventory, risks/gaps list.

****Acceptance:**** Every claim in this document is verified or corrected. No code behavior changes.

Branch 1 â€” Canonical evidence contract

****Goal:**** Create one stable interface for all existing modules.

****Steps:**** Add Pydantic/typed models; adapters for infra, identity and MiniLM; compatibility adapter for legacy fusion input; schema validation tests; unique IDs and idempotency keys.

****Data flow:**** Existing module output -> adapter -> `EvidenceUnit` validation -> PostgreSQL canonical evidence store -> fusion.

****Acceptance:**** Existing signals operate through the new schema without regression; malformed evidence is rejected/audited.

Branch 2 â€” Capture provenance and safe collection hardening

****Goal:**** Make collection safe, traceable and resilient.

****Steps:**** Add capture records, MinIO/fixture hash storage, policy allowlist/blocklist, authorization expiry, status events, Tor egress enforcement, timeouts, response caps, MIME validation, quarantine statuses and retry logic.

****Data flow:**** Source policy -> collection job -> capture/raw artifact -> normalization/quarantine -> evidence-ready document.

****Acceptance:**** Offline/unavailable, mirror/repost, oversized and blocked-source fixtures behave as specified; old evidence remains available after source failure.

Branch 3 â€” Explainable fusion and link lifecycle

****Goal:**** Replace bare linkage score with versioned, explainable candidate association.

****Steps:**** Create `config/scoring.yaml`; classify K/I/B/S; noisy-OR/group duplicate evidence; enforce category caps; tier mapping/hysteresis; conflict sets; reversible associations; deterministic explanation templates; versioned rescore events.

****Data flow:**** Valid evidence -> pair/group/independence set -> category scores -> `CandidateLink` -> graph/timeline/audit projection.

****Acceptance:**** Easy, hard, negative, stale, duplicate and conflict fixture tests pass; no API returns only a raw score.

Branch 4 â€” OnionScan wrapper adapter

****Goal:**** Add broader OPSEC observations with isolated upstream dependency.

****Steps:**** Pin OnionScan artifact; invoke in a non-root restricted subprocess/container; validate target against allowlist; timeout/output cap; save raw output/hash; parse known JSON/structured fields; generate infrastructure evidence and caveats; fixture replay fallback.

****Data flow:**** Approved target -> subprocess -> raw result artifact -> adapter -> `EvidenceUnit` -> I category fusion.

****Acceptance:**** Mock/raw fixture parses correctly; timeout/schema-error fixture creates safe status and does not break pipeline.

Branch 5 â€” Dashboard evidence and analyst review

****Goal:**** Add evidence-first UI without rebuilding current dashboard.

****Steps:**** Build profile summary, graph edge drawer, explanation view, limitations, redaction, decision dialog, role checks and audit events. Add graph depth/permission filtering.

****Data flow:**** Dashboard -> link/profile API -> redacted CandidateLink/EvidenceUnit -> analyst decision -> new audit + link version.

****Acceptance:**** Every visible edge opens evidence chain; decisions require reasons; viewer access is restricted.

Branch 6 â€” Timeline

****Goal:**** Enable date-bounded investigation and transparent history.

****Steps:**** Build TimelineEvent generator, observed/captured time display, confidence changes, source availability events, decision events and date filters shared with graph.

****Data flow:**** Evidence/link/decision history -> timeline builder -> timeline API -> UI -> evidence drawer.

****Acceptance:**** Demo includes at least three event types, uncertain source time is visibly marked and date filters affect both graph/timeline.

Branch 7 â€” Behavioral and persona migration signals

****Goal:**** Detect plausible rebranding with multiple weak signals.

****Steps:**** Build posting-time histograms, vocabulary overlap, rare-template hashes, source/category overlap; require a non-name contextual condition; emit `persona\_migration\_candidate`; create conflict-aware candidate links.

****Data flow:**** Alias posts -> feature builder -> B evidence -> fusion -> candidate link/timeline.

****Acceptance:**** `GhostVendor -> Nightshade99` test reaches defensible review tier; copied template and similar-name negatives do not.

Branch 8 â€” Classical stylometry (optional but recommended)

****Goal:**** Add independent writing-style evidence.

****Steps:**** Implement eligibility gates; clean corpus; compute classical features with `pystylometry`/scikit-learn; calibration fixtures; confidence cap; model/corpus metadata.

****Data flow:**** Eligible alias corpora -> classical analyzer -> S evidence -> fusion.

****Acceptance:**** True/negative fixtures order correctly; code-switched/short/LLM-like/template-only samples emit no or low-quality excluded evidence; no text-only high tier.

Branch 9 â€” Case export, audit and governance

****Goal:**** Deliver reproducible reports and access controls.

****Steps:**** Implement cases, append-only/tamper-evident audit strategy, RBAC, redaction policy, export snapshot transaction, JSON/CSV/PDF templates, retention/legal hold/tombstone process.

****Data flow:**** Case selection -> authorization -> snapshot exact IDs/versions -> report artifact -> export audit event.

****Acceptance:**** Changing evidence during export does not alter generated report; unauthorized user cannot export protected data; report contains caveats and metadata.

Branch 10 â€” Neo4j migration (stretch)

****Goal:**** Improve time-bounded graph traversal after core system is stable.

****Steps:**** Introduce graph repository interface; backfill canonical entities/evidence/links; dual-write and compare outputs; add migration rollback; query by actor/date/depth; preserve all evidence IDs on edges.

****Data flow:**** Canonical PostgreSQL records -> graph projection writer -> Neo4j -> graph API. PostgreSQL remains source of truth.

****Acceptance:**** Date-bounded multi-hop query works and projection reconciliation finds no lost evidence/links.

Branch 11 â€” Demo hardening

****Goal:**** Produce a dependable, safe SIH demonstration.

****Steps:**** Offline fixture pack, `make demo-reset`, preconfigured users, no live URLs/illegal content, CI checks, 5-minute script, negative-case rehearsal, fallback screenshots/export.

****Data flow:**** Fixture replay -> all adapters -> fusion -> dashboard -> case/export.

****Acceptance:**** Fresh Docker setup runs without internet; all four demo scenarios complete in sequence.

Part G â€” Test, Deployment and Demo

23. Required test fixtures

1. Easy: shared PGP in two profiles.
2. Hard: text/behavior-only pair -> no higher than Possible Association.
3. Rebrand: GhostVendor -> Nightshade99.
4. Negative: shared generic/mixer/escrow-style wallet or copied template.
5. Offline source -> 503 -> changed content.
6. Mirrored/reposted identical page.
7. Oversized/malformed HTML and invalid encoding.
8. Published PGP block vs verified signature.
9. Unicode-confusable/recycled alias.
10. Competing actor-hypothesis conflict set.
11. Translation/code-switching/short corpus.
12. LLM-like/intentional style imitation.
13. Stale certificate/header artifact.
14. Concurrent duplicate collection jobs.
15. Decision/evidence change during export.
16. Redaction and unauthorized export attempt.

24. Offline demo script

1. Start with `make demo-reset && docker compose up --build`.
2. Log in as Analyst.
3. Search `GhostVendor`.
4. Show shared PGP/wallet evidence and tiered technical association.
5. Open edge evidence drawer; point out proof, confidence breakdown and caveat.
6. Open timeline to show market migration/rebrand.
7. Show the hard text-based case and explain why it remains only Possible Association.
8. Show the negative shared-wallet/template case and its caveat/low tier.
9. Accept or mark a link for review with an analyst note.
10. Export report and show evidence IDs, score model version and disclaimer.

25. Deployment checklist

- All containers start from pinned versions; no runtime model downloads.
- Model artifacts and fixture data are bundled with checksums.
- `.env` is excluded from Git; secrets scanner passes.
- Collector direct egress denied; configured Tor proxy health check passes.
- Default mode is `fixture\_replay`; live mode must be explicitly enabled by admin policy.
- Backups/recovery documented for PostgreSQL and MinIO.
- SBOM and third-party notices available.

26. Jury Q&A answers

****Is this real de-anonymization?**** No. It is confidence-scored technical attribution assistance for authorized analysts. It creates auditable leads, not proof of a real-world person.

****Did you build all scanning logic?**** We retained and extended our native certificate correlation, and integrate a pinned open-source OnionScan dependency through an adapter for additional OPSEC observations. Our contribution is the evidence model, fusion, analyst review, timeline and explainability layer.

****Why not use Maltego or SpiderFoot?**** Those are broad OSINT platforms. This product is deliberately narrow: dark-web attribution using evidence chains, time-aware actor links and controlled technical signals.

****How is this different from commercial threat-intelligence vendors?**** Commercial vendors win on proprietary data coverage and analyst operations. Our prototype demonstrates a self-hostable, evidence-first correlation layer that shows why it made a technical association and can operate with controlled/offline data.

Part H – Required Repository Documents

Create these files before completion:

```
```text
docs/PRD_FINAL.md
/docs/APP_DATA_FLOW_FINAL.md
/docs/EDGE_CASE_REGISTER.md
/docs/baseline-audit.md
/docs/architecture.md
/docs/data-contracts.md
/docs/scoring-methodology.md
/docs/threat-model.md
/docs/data-governance.md
/docs/third-party-notices.md
/docs/runbook-local-demo.md
/docs/analyst-guide.md
/docs/test-plan.md
/docs/demo-script.md
/docs/jury-qa.md
/config/scoring.yaml
/fixtures/README.md
```
```

27. First instruction for a human or agent

```
```text
Use this Final Build Guide as authoritative. Begin Branch 0 only. Inspect the repository without modifying code. Create docs/baseline-audit.md with: module path, entry point, input/output, current dependencies, test/fixture coverage, data-provenance gaps, security gaps and exact adapter required for EvidenceUnit compliance. Do not add new tools or refactor existing modules until the audit is reviewed and accepted.
```
```